

# Consent-O-Matic: Automatically Answering Consent Pop-ups Using Adversarial Interoperability

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## ABSTRACT

The majority of consent pop-ups on the web do not meet the requirements for legally valid consent laid out in the General Data Protection Regulation (GDPR). In the face of a lack of enforcement, we present the browser extension *Consent-O-Matic* which uses *adversarial interoperability* to automatically answer these pop-ups based on the user's preferences. We document how the current implementation of these pop-ups support and inhibit interoperability, focussing on the difference between static and dynamic HTML, the quality of the semantic markup, and the visibility of the system's state; and we present the implementation of *Consent-O-Matic*. Lastly, we discuss the possibilities, limitations, and concerns of an adversarial approach.

## CCS CONCEPTS

• Security and privacy → Human and societal aspects of security and privacy; • Applied computing → Law, social and behavioral sciences.

## KEYWORDS

General Data Protection Regulation, regulatory enforcement, adversarial interoperability, browser extension

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## 1 INTRODUCTION

The majority of consent pop-ups on the web do not meet the requirements for legally valid consent laid out in the General Data Protection Regulation (GDPR) [12, 18]. The lack of effective enforcement of the GDPR is one of the core criticisms that the regulation has faced after going into effect in 2018 [10], and has also been flagged by the European Commission's own internal assessment in 2020 [7]. Explanations for this lack of enforcement are disputed, ranging from insufficient resources for the national data protection authorities [2], the "one-stop-shop" mechanism that is used to handle violations across national borders [1], and accusations of industry capture and corruption [11].

Regardless of the reason for a lack of enforcement of the GDPR in general, and that of consent pop-up designs on the web in particular, the consequences are that users' rights to data protection are consistently violated while the slow-moving regulatory apparatus tries to correct course. Although we do not subscribe to the idea that systemic issues can be "solved" by technological innovations (i.e., techno-solutionism [14]) and encourage a multi-dimensional response to the lack of regulatory compliance, researchers and non-profits have shown that it is possible to use technology in order to effectively resist other technology (see, e.g., Irani and Silberman [8], Merzdovnik et al. [13]).

We present *Consent-O-Matic*, a browser extension that automatically answers consent pop-ups on the web [15]. *Consent-O-Matic* allows people to set their data processing preferences once and let the extension do the rest, bypassing the (non-compliant) interfaces of existing consent pop-ups. The extension is built using the idea of *adversarial interoperability* [5]—interfacing with other systems without their permission or support. As a design approach, this has implications for both the process of system development and the effectiveness and robustness of the outcome. However, *Consent-O-Matic* is currently empowering more than 16,000 active users [16, 17], demonstrating that these conditions can lead to useful systems and, in this, case alleviate regulatory failures.

The paper is structured as follows. First, we outline related work that uses automated systems for privacy and data protection, and conceptual discussions of interoperability. Second, we describe the general design of consent management platforms, the specific

technical implementation of the five most popular third-party pop-up services, and how these support or inhibit interoperability. Third, we explain how we built our extension to take advantage of and deal with these conditions.

## 2 RELATED WORK

### 2.1 Automating Consent

Trying to automate how users communicate and enforce their data processing preferences is not new. Two of the most high-profile attempts to build a way for users to do this directly into the web's standards are the *Privacy Preferences Platform* (P3P) [19] and *Do Not Track* (DNT) [20], both of which were proposed by the World Wide Web Consortium (W3C)<sup>1</sup>. P3P is a protocol that allows websites to express what data they collect in a machine-readable format, which can then be compared to a user's preferences sent in the header of an HTTP request, and automatically applied when the website is served. DNT is a field in an HTTP header whose binary value (true or false) communicates to a webserver whether the user wants to be tracked. Both of these standards-based attempts were ultimately unsuccessful, mainly because they relied on self-regulation, i.e., on stakeholders such as advertising agencies and website publishers to adapt their systems and respect the signals they received from users. In the wake of changes in global data protection regulation (specifically the California Consumer Privacy Act), a diverse group of organisations have recently proposed a spiritual successor to DNT—Global Privacy Control<sup>2</sup>—but whether it will be more successful remains to be seen.

Privacy organisations and entrepreneurial web users have used a more bottom-up, adversarial approach to (re)gain control over their data by building browser extensions that intervene at the infrastructural level of data collection. Two of the most popular ones are *Ghostery* [4] and *PrivacyBadger* [6], both of which block network traffic between a user and trackers. *Ghostery* does this based on a human-curated blacklist of known tracker endpoints, whereas *PrivacyBadger* uses an algorithmic approach that detects (and remembers) suspicious network behaviour.

More recently, browser extensions have appeared that intervene on the interface level of privacy notices. These extensions were motivated by the response of websites to new European regulations, in particular the ePrivacy Directive (2002) and the General Data Protection Regulation (2018). The former law requires a website to have consent from a user before they can read or write data on their “terminal equipment”, and the latter requires anyone dealing with personal data to have a legal basis for processing it. Actors in the online advertising ecosystem, such as website publishers and adtech vendors, responded by designing and implementing consent pop-ups—so-called Consent Management Platforms—which have become a nuisance for many users of the web. The browser extensions developed in response to these laws predominately focus on improving the users' experience of the website by removing the notifications, but do not necessarily improve the user's control over their personal data. For example, *Cookie Notice Blocker* (20.000+

users on Chrome)<sup>3</sup> and *Consent Manager* (8.000+ users)<sup>4</sup> both inject JavaScript to block the notices from appearing. However, many of these notices use an opt-out approach, which means removing HTML elements does not protect against tracking. We only found one example of a tool that tries to improve the user experience of consent pop-ups but also improve the user's control over the data that is collected on them by interacting with the pop-ups: *autoconsent*, developed by Sam Macbeth for the (now discontinued) privacy-oriented Cliqz browser<sup>5</sup>.

*Consent-O-Matic* combines elements of these three approaches: it automates the interaction with consent pop-ups to improve the experience of the web, and it gives users control over their data by basing those interactions on the privacy preferences they set.

### 2.2 Interoperability

Interoperability refers to “the ability of a system, product or service to communicate and function with other (technically different) systems, products or services” [9]. Interoperability has become a popular topic in (European) technology policy debates, primarily in the context of competition regulation. In these discussions, interoperability is seen as a mechanism to address failures in digital markets, which have a tendency to “tip” (i.e., a single actor capturing most of the market share and becoming entrenched) [3]. By forcing companies to allow competitors to interoperate with their systems, network effects are weakened and it is easier for new actors to enter the market and provide more choice for consumers. When proposing possible interoperability obligations, the concept is often broken down into horizontal and vertical interoperability, with the former referring to interoperability between competing services (e.g., sending messages from WhatsApp to Signal), and the latter between complementary services (e.g., having access to multiple app stores on iOS devices) [9].

Cory Doctorow, writing for the Electronic Frontier Foundation, has published a collection of articles conceptualising and historicising interoperability in the tech industry (see Doctorow [5] for an overview). In these, he divides interoperability into separate constructs not based on technical functioning, but on intent:

- *Cooperative interoperability*: collaborative efforts by multiple systems to work together (e.g., the Semantic Web).
- *Indifferent interoperability*: a neutral attitude, neither facilitating nor inhibiting exchange.
- *Adversarial interoperability*: the efforts of one system to interface with another against the explicit wishes of its progenitors (e.g., Apple's reverse engineering of Microsoft's .doc format so Pages could open them).

Interoperability – whether conceptually, technologically, or empirically – does not appear to be a key topic of interest for HCI research. Querying for the term in the entire corpora of CHI, UIST, CSCW, GROUP, ECSCW, TOCHI, and JCSCW returns 339 results.

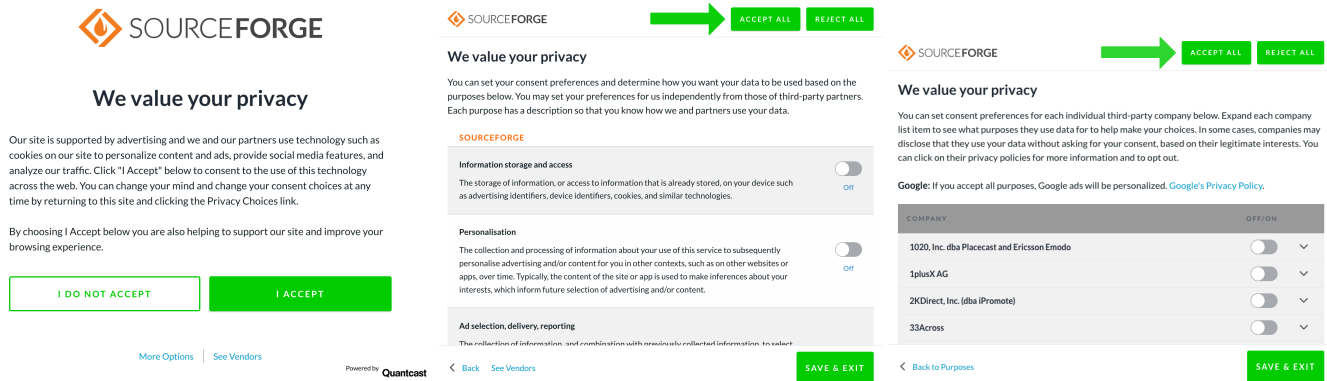
<sup>1</sup>The international organisation that is responsible for maintaining and developing the standards for the World Wide Web.

<sup>2</sup><https://globalprivacycontrol.github.io/gpc-spec/>

<sup>3</sup><https://chrome.google.com/webstore/detail/cookie-notice-blocker/odhmfimnoejhihkmfebnoljiibpnednn?hl=en-GB>

<sup>4</sup><https://chrome.google.com/webstore/detail/consent-manager/gpkaoajillfmplnglbaggplnphadbfbah?hl=en-GB>

<sup>5</sup><https://github.com/cliqz-oss/autoconsent>



**Figure 1: The Consent Management Platform design of Quantcast. The three layers provide the user with increasingly more granular consent options, going from bulk consent, to purpose-level, and lastly vendor-level consent.**

Filtering for only those publications with “interoperability”, “interoperable”, or “interoperating” in the title, abstract, or author keywords (and removing workshop papers and false positives) reduces that to a mere 30. Almost half mention interoperability tangentially, either as the problem statement (6) or as an implication for design (6). Twelve papers present a system which includes interoperability to varying degrees (i.e., as its main feature, or as an unintentional side-effect). Four papers discuss interoperability conceptually, all in the context of supporting collaboration and cooperation.

In this paper, we use Doctorow’s concept of *adversarial interoperability*, not to increase competition, but to empower users and address the lack of regulatory enforcement. By doing this, we conceptually develop adversarial interoperability as a design approach and reflect on the opportunities and challenges it creates.

### 3 CONSENT MANAGEMENT PLATFORM DESIGNS

Consent Management Platforms (CMPs) are (third party) services that websites use to collect consent from their visitors in order to read and write data onto their machines (e.g., cookies) and to process their personal data. Often, websites use other services to automatically place advertising on their page, so many CMPs also provide scanning capabilities to detect whether data is collected for new purposes or vendors that require the user re-consent. In Q3 2020, 58% of the 10,000 most popular websites in the UK<sup>6</sup> used one of five third-party CMPs: QuantCast, OneTrust, TrustArc, Cookiebot, or Crownpeak.

Visually, the CMP interface generally has three parts: the bulk yes-or-no consent option on the first page, a second page with specific purposes of data processing, and a third page with third-party vendors for whom data is collected and with whom it is shared (see Fig. 1).

Operationally, the CMPs go through a number of basic steps. When a user accesses a site, the CMP detects their IP address to see if they fall within the territorial scope of the GDPR. If they do, the CMP checks the user’s device for any previously set consent

preferences in cookies or local storage, and retrieves this data. If this fails, or if the CMP decides the user’s preferences have expired, they are shown a consent notice. Based on the user’s interaction with the pop-up or the page, and the type of actions the CMP counts as an answer (e.g., clicking a button, or closing the window), a response is recorded and stored in a cookie. This consent status is then communicated to any advertising services the websites uses for real-time ad bidding.

## 4 DESIGNING FOR ADVERSARIAL INTEROPERABILITY

Using an adversarial interoperability approach when building a system has implications for the design process and the design outcome. We will explain how this constraint affected these two aspects in the following sections.

### 4.1 Design process: reverse engineering Consent Management Platforms

Interoperating with another system without their permission or support means that there are no standards or APIs that can be used or relied upon. Instead, we gathered information about the implementation of the CMPs by reverse engineering the HTML, CSS, and JavaScript, by looking at the documentation the CMPs made available for their clients, and by reaching out to the CMP providers as a potential customer. We found that, although most CMPs look and function very similarly from the user’s perspective, their technical implementations vary wildly. Below we will explain the main differences and their implications for interoperability.

**4.1.1 Dynamic vs. static HTML.** Most CMPs load the content of the pop-up dynamically as the user interacts with the interface – only Cookiebot is entirely present in the document object model (DOM) on pageload. This makes a big difference when trying to interoperate with the CMP, because it requires a mechanism that continuously checks whether elements are present on the page and handles the time between those checks. This puts an upper limit to how fast we can automate interaction with the interface, because after each action we need to wait and check whether the CMP has

<sup>6</sup>According to Adzerk, a company that does server-side ad serving and writes reports about the state of the industry: [www.adzerk.com](http://www.adzerk.com)

finished loading its content. TrustArc, for example, makes a lot of remote calls to retrieve information about which vendors are on the website, with delays sometimes lasting thirty seconds.

Static HTML – or more precisely, HTML which is static after the page finishes loading – makes it more straight-forward to interoperate with, because all elements are present in the DOM. A unique benefit of this design for interoperability is that the adversarial system might have access to more functionality than the user. Cookiebot, for example, has a “Decline all” button in their HTML, but makes it possible for website owners to hide this button from the user. However, because it is present in the DOM, it can be clicked programmatically, which means the adversarial system could outperform even the best user, just by virtue of being able to select options that are never visually represented.

**4.1.2 Semantic markup.** A key part of interoperating with another system is being able to accurately and reliably target objects of interest. Semantic information that describes the object’s meaning is one key way to facilitate this. The target elements in these CMPs are the buttons, sliders, checkboxes, and toggles that set and submit the consent preferences. In HTML, explicit semantics are added using tags, classes, ids, and attributes.

The quality of the semantic markup varies significantly between different CMPs. Cookiebot has very descriptive ids for their checkboxes (e.g., `CybotCookiebotDialogBodyLevelButtonMarketing`), while QuantCast, Crownpeak, OneTrust, and TrustArc use identical class names for all the purposes (e.g., `category-check`), making it difficult to distinguish between them. Certain versions of OneTrust even have factually incorrect semantics: purposes and vendors are placed under the `menu-item-necessary` category, yet are in fact optional. Some CMPs use auto-generated classes, making it impossible to consistently interoperate with it.

If no semantic markup is available, a specific element can be targeted by traversing the DOM tree, but this is not very robust because any changes to the DOM structure would break the detection. If semantic information in the code is not possible, the only other option is to analyse the text content of elements to understand what it represents or controls, but this creates problems when website owners customise button labels, when there are typos, or when an unknown language is used.

**4.1.3 Hidden state.** Proper software engineering principles argues for encapsulation of data within methods that use it, and against storing state in something such as the DOM. Indeed, most of the CMPs store the state of the user’s consent entirely in the JavaScript runtime state. However, this makes it difficult for an adversarial system to verify whether its interactions with the other system actually change the consent signal that is submitted. For example, in HTML the state of a checkbox element is represented by the attribute `checked` on that element. However, this does not mean that changing that attribute programmatically will be interpreted by the CMP as the user checking or unchecking the checkbox; it might instead be using a custom `EventListener` on the checkbox to determine whether a user is agreeing or rejecting to some data processing. As a result, interoperating with a system where the state is hidden means making direct changes to the data is unreliable and instead requires simulating user interactions.

## 4.2 Design outcome: the *Consent-O-Matic* browser extension

Once we know how the systems that we are trying to interoperate with are designed, we are able to build our adversarial response. Because of this design approach, our implementation is constrained by two concerns: 1) we need to work with the design decisions of the other systems, which are naturally not optimised for interoperability, and; 2) there is no guarantee that these designs are stable, so our implementation should balance accuracy with robustness. The extension is available on the Chrome [16] and Firefox [17] webstore and the open source code on Github [15].

*Consent-O-Matic* goes through two steps whenever a user visits a website. First, it detects whether a CMP is present in the DOM and present visually on the page (see Figure 2c). Second, if it has detected a CMP is present, it executes a set of actions to set the CMP according to the user’s preferences and submit the consent (see Figure 2d-2f). The information necessary to be able to execute these operations for a particular CMP are stored in a set of JSON rules, which makes it easy to extend the extension with new CMPs or edit existing instructions (see Figure 2a for an example).

**4.2.1 DOM selection.** To detect elements, we built a DOM selection mechanism that is more sophisticated than the built-in methods in JavaScript. The *DOM selection* works by first using a CSS selector to target elements and subsequently applying (optional) filters to what is returned to further winnow down the list (see Table 1 for the five types of filters).

Because the way some CMPs are implemented makes it very difficult to target certain elements, it is also possible to execute this selection process in two steps using the parent and target construction. This allows users to first select the “root” parent node and subsequently select one or multiple targets relative to that node.

**4.2.2 Actions.** Actions are what the *Consent-O-Matic* extension does once a target has been selected, which mimic user interactions with the interface. Some actions are for executing something to a target element, other actions are about controlling the flow of interactions with the CMP (see Table 2 for the nine types of actions). These actions are adjusted based on the consent preferences submitted by the user.

**4.2.3 Consent Preferences.** The user can specify their privacy preferences by toggling six different data processing purposes in the extension’s settings page. These purposes are based on a survey of the purposes found in 680 CMPs used by the top 10,000 most popular websites in the UK [18]. Different spellings of the purpose category names were first collapsed by using OpenRefine<sup>7</sup>. Then, the descriptions of those different purposes were compared and further merged until six separate ones remained: Information Storage and Access; Preferences and Functionality; Performance and Analytics; Content selection, delivery, and reporting; Ad selection, delivery, and reporting; and Other Purposes. The purposes found in the CMPs we interoperate with are matched to these global data processing categories and the user’s answer for each of them (opt-in or opt-out) are passed to the consent action mentioned previously.

<sup>7</sup>An application used for data wrangling that has various fuzzy matching algorithms.

```

1 {"CMP4U": {
2   "detectors": [
3     {"presentMatcher": {
4       "type": "css",
5       "target": {
6         "selector": "#theCMP"}}}],
7   "methods": [
8     {"name": "OPEN_OPTIONS",
9      "action": {
10        "type": "click",
11        "target": {
12          "selector": ".options-button",
13          "textFilter": "change settings"}},
14     {"name": "DO_CONSENT",
15      "action": {
16        "type": "consent",
17        "consents": [
18          {"type": "A",
19           "matcher": {
20             "type": "checkbox",
21             "target": {
22               "selector": ".vendor-item",
23               "textFilter": "Analytics"}},
24           "toggleAction": {
25             "type": "click",
26             "target": {
27               "selector": ".vendor-item",
28               "textFilter": "Analytics"}}}}],
29     {"name": "SAVE_CONSENT",
30      "action": {
31        "type": "click",
32        "target": {
33          "selector": ".save-consent-btn"}}}]]}}

```

(a) JSON ruleset for interacting with the CMP “CMP4U”.

**Consent-O-Matic**  
Your Choice - Applied Everywhere

Your Choice Rule Lists Log

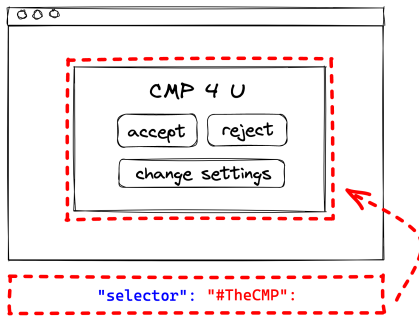
Configure what categories of tracking you would like to allow. Consent-O-Matic then fills out consent popups as best as possible based on your preferences. All tracking is rejected by default.

- ☐ **Information Storage and Access**  
Storage of information or access to information that is already stored on your device - such as advertising identifiers, device identifiers, cookies, and similar technologies.
- ☐ **Preferences and Functionality**  
Allow sites to remember choices you make (such as your user name, language or the region you are located in) and provide enhanced, more personal features. For instance, these cookies can be used to remember your login details, changes you have made to text size, fonts and other parts of web pages that you can customize. They may also be used to provide services you have asked for such as watching a video or commenting on a blog. The information in these cookies is not used to track your browsing activity on other websites.
- ☒ **Performance and Analytics**  
The collection of information, and combination with previously collected information, to measure, understand, and report, on your usage of the services. This allows websites to count visits and traffic sources so they can measure and improve the performance of the site. It helps them know which pages are the most and least popular, see how visitors move around the site, and where visitors come from.
- ☐ **Content selection, delivery, and reporting**  
Collection of information, and combination with previously collected information, to select and deliver content for you, and to measure the delivery and effectiveness of such content. This includes using previously collected information about your interests to select content, processing data about what content was shown, how often or how long it was shown, when and where it was shown, and whether you took any action related to the content, including for example clicking on content. The data will be used to personalise content on the website itself, but also in other contexts such as other websites, apps, browsers, and devices.
- ☐ **Ad selection, delivery, and reporting**  
Collection of information, and combination with previously collected information, to select and deliver advertisements, and to measure the delivery and effectiveness of such advertisements. This includes using previously collected information about your interests to select ads, processing data about what advertisements were shown, how often they were shown, when and where they were shown, and whether you took any action related to the advertisement, including for example clicking an ad or making a purchase. The data will be used to personalise advertising on the website itself, but also in other contexts such as other websites, apps, browsers, and devices.

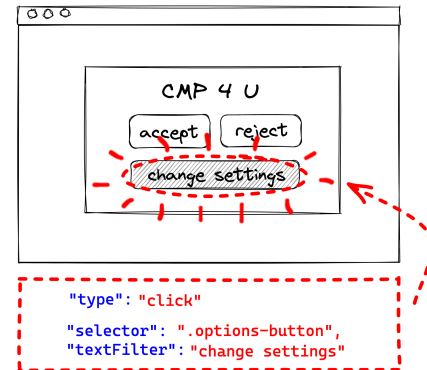
Also includes:  
Google

- ☐ **Other Purposes**  
Unclassified data collection for which the purpose is not clearly described by the website or where the data collection and processing does not fit any other category

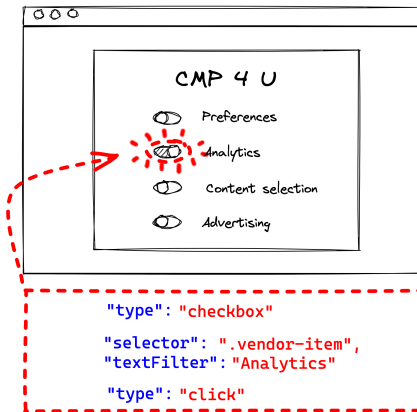
(b) Consent-O-Matic preference center.



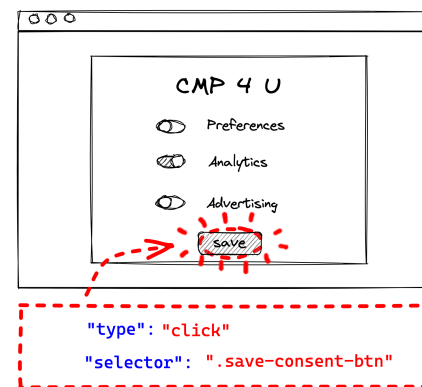
(c) Detecting CMP presence



(d) Opening purpose-level consent settings.



(e) Toggling “Analytics” purpose.



(f) Saving consent preferences.

Figure 2: Figure 2a and 2b show the two core components of *Consent-O-Matic*: a ruleset for a CMP and a user’s preferences. Figure 2c-2f show how *Consent-O-Matic* interacts with a CMP on a website.

**Table 1: The different filtering mechanisms that can be used to select a particular DOM element**

| Filter type   | Result                                                                                                                                                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| textFilter    | filters all nodes that do not include the given (array of) string(s)                                                                                     |
| styleFilter   | filters based on computedStyle. Takes an option which is the CSS property name, and a value. Can be negated True or False to specify presence or absence |
| displayFilter | filters based on if the element is displayed or not                                                                                                      |
| iframeFilter  | filters based on if a node is inside an iframe or not (Boolean)                                                                                          |
| childFilter   | filters based on whether the node has a particular child element, which is specified using the same DOM selection process                                |

**Table 2: The different actions that can be performed on an element to simulate a user's interaction**

| Action  | Result                                                                                                    |
|---------|-----------------------------------------------------------------------------------------------------------|
| click   | simulates a click                                                                                         |
| slide   | simulates moving a slider                                                                                 |
| hide    | adds display & none                                                                                       |
| close   | closes the current tab (for when new ones are opened)                                                     |
| ifcss   | conditionally executes an action based on whether an element is or is not found.                          |
| waitcss | waits until it can or cannot find the target element after a specified waitTime for the number of retries |
| foreach | runs the same action for each of the elements found                                                       |
| consent | takes a list of the user's consent choices (see below) and tries to toggle an element value based on that |
| list    | takes a list of actions to apply to the same element.                                                     |

## 5 DISCUSSION AND CONCLUSION

*Consent-O-Matic* was first launched in December 2019, and is currently being used by more than 16.000 users. This demonstrates that building systems using adversarial interoperability can be useful to empower users, and robust enough to still operate after more than two years with minimal maintenance.

Future work will focus on covering more CMPs, and to include more languages in the text filtering mechanism. Now that Apple has released their WebExtensions API to align with Chrome and Firefox, we will also make *Consent-O-Matic* available for macOS and iOS users. User evaluations of *Consent-O-Matic* could focus on whether automatically answering consent pop-ups results in better alignment between a user's preferences and their consent signal compared to answering it manually. The results of such a study would be valuable to the ongoing discussions around the ePrivacy Regulation, the European Commission's "sister" proposal to the GDPR that will revamp the rules about reading and writing on people's devices. One contentious element of that proposal is whether consent signals from a browser are valid, which if approved could make signals from interventions such as *Consent-O-Matic* or DNT legally enforceable.

While building adversarial systems could be used to empower users, we are also aware that we are now acting as a regulatory intermediary and enforcing the GDPR in a particular way. By installing *Consent-O-Matic*, users have given us considerable control and responsibility over how their data is handled, and there is no guarantee that we are necessarily acting on their behalf more or less than the CMP providers. Interoperability, ultimately, is about power. With increasing monopolisation of (commercial) technology, facilitated by walled-garden designs and network effects, advancing

the concept of interoperability is an agenda primarily interested in redistributing power, recasting tech companies from philosopher kings to democratic equals. This requires more projects that explore what technological characteristics can support interoperability, and policy discussions about how to incentivise and protect the implementation of those characteristics.

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